More Laws, More Growth? Evidence from US States

Elliott Ash, Massimo Morelli, and Matia Vannoni

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Motivation & Research Question

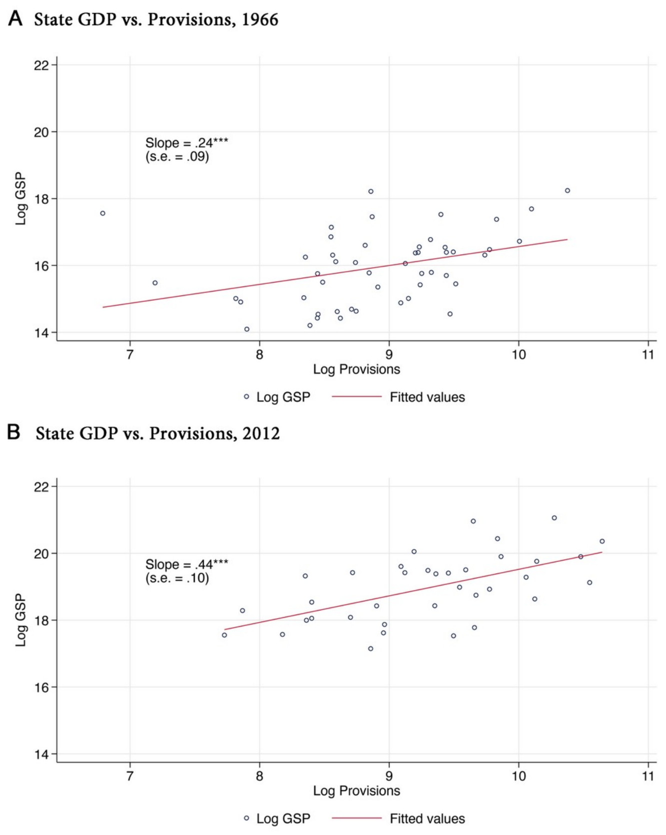

FIG. 1.—State GDP and legislative output, 1966 and 2012. Scatterplots show the relationship between (log) provisions and (log) state GDP at the beginning of our time period (1966; *A*) and the end (2012; *B*).

Note: Log state GDP vs. log legal provisions across states, at the start (1966; A) and end (2012; B) of the sample. The positive cross-sectional slope motivates the causal question. (Figure 1)

- In the cross section, states with larger, more complex legal systems also have larger, more productive economies (see figure).
- But is this causal? A bigger economy may mechanically require more laws (reverse causality) and shocks may drive both.
- Positive view: more complete 'legislative contracts' aid enforcement and investment. Negative view: excess lawmaking raises compliance costs or serves special interests.
- Question: does more legislative detail CAUSE growth in US states, 1965-2012, and through what mechanism?
- Preview: at the margin, more legislation raises growth.

Data

- Panel of the 50 US states by biennium (2-year periods), 1965-2012; states publish compiled statutes about every 2 years.
- Outcome: growth = change in log real per capita gross state product (GSP), from BEA Regional Accounts; also GDP, population, employment, income, profits, establishments.
- Legislation: full text of US state session laws (all statutes enacted each session), digitized via OCR (high quality; about 3% misspelling rate).
- Controls: demographics (census), state government finances, and political control (number of Democratic-held bodies).
- Mechanism data: Nunn (2007) relationship-specificity for 30 NAICS sectors; local economic policy uncertainty (EPU) built from newspaper text.

Measuring Legislation from Statute Text (NLP)

- Segment each biennium's corpus into statutes (roughly a bill or code chapter), then into sentences.
- Apply a syntactic dependency parser and extract 'provisions' = legally relevant statements (delegations, constraints, permissions), per Vannoni, Ash & Morelli (2019).
  - e.g., a constraint = 'the Agent shall not' / 'is not allowed' / 'is prohibited from'.
- Legislative output W(s,t) = count of legal provisions in state s, biennium t (in logs) -- cleaner than word or page counts, which include non-legislative text.
- Allocate statutes to legal topics with an LDA topic model (Blei et al. 2003); tuned to K=18 (and K=42) topics.
  - Topic flow W^k(s,t) = provisions weighted by each statute's topic share -- the building block of the instrument.
- Contingency lexicon: a provision is contingent if it contains {if, unless, where, provided that, ...}; about 1/5 are contingent. Contingencies split the state space and cut enforcement ambiguity.

The 18 Legal Topics (LDA)

TABLE 1
List of Topics, 18-Topic Specification

Label	Frequency	Most Associated Words
Courts	.0724	court, judgment, attorney, case, appeal, civil, petition, sheriff, trial, circuit court, district court, such person, complaint, counsel, brought, circuit, warrant, paid
Pensions	.0653	paid, benefit, rate, payment, equal, death, age, credit, pay, total, life, pension, premium, calendar year, loss, account, case, per cent, event, membership, excess, maximum
Local projects	.0645	development, local, project, budget, government, cost, grant, research, center, local government, data, transfer, governor, is the intent, develop, urban, review, biennium
Procurement	.0621	director, contract, work, review, civil, labor, contractor, attorney general, bureau, final, perform, audit, receipt, status, exempt, panel, government, firm, bid, prepared
Elections	.0612	district, town, petition, charter, special, ballot, mayor, voter, township, precinct, cast, referendum, census, elector, case, town council, said district, such district
Banking	.0604	loan, trust, bank, agent, partnership, institution, foreign, stock, mortgage, deposit, surplus, interest, merger, credit union, partner, case, credit, gift, branch, transact
Licensing	.0593	license, fee, dealer, sale, food, sold, holder, sell, valid, fish, agent, distributor, milk, liquor, product, such license, livestock, game, card, retail, misdemeanor, fine
Real estate	.0576	real, interest, sale, owner, contract, claim, lien, payment, transfer, instrument, seller, holder, issuer, debtor, claimant, buyer, pay, broker, settlement, receipt, money
Bonds	.0574	interest, bond, payment, commonwealth, cost, sale, paid, pay, project, power, thereon, sold, debt, pledge, local law, event, hereof, proper, said board, real, port, sell, therefrom
Expenditures	.0569	fund, account, money, paid, special, pay, tile, payment, transfer, for the fiscal year, excess, trust fund, so much thereof, deposit, state general fund, auditor, tie
Bureaucracy	.0551	governor, council, government, chief, fire, appoint, personnel, compact, conflict, perform, shall consist, invalid, parish, successor, volunteer, membership, head, travel
Healthcare	.0546	health, care, treatment, health care, physician, home, human, patient, mental, mental health, drug, social, condition, public health, medicaid, dental, client, review, institution
Child custody	.0535	child, court, minor, children, parent, age, probation, crime, victim, parole, guardian, adult, petition, placement, youth, case, social, legal, child support, obligor, home
Taxes	.0522	tax, paid, gross, credit, return, net, rate, exempt, assessor, case, refund, equal, sale, total, calendar year, payment, fuel, portion, sold, price, retail, zone, pay, such tax
Land and energy	.0512	land, water, owner, control, site, air, solid, gas, tenant, oil, park, airport, forest, coal, plant, environment, prevent, underground, power, soil, portion, landlord, condition
Education	.0474	school, school district, state board, district, student, institution, higher, teacher, special, aid, pupil, children, school year, tuition, high school, school board

Note: The 18 LDA topics from state session laws, by frequency, with most-associated words. Topics are interpretable policy areas (courts, pensions, licensing, taxes, land/energy, …) and no single topic dominates. Pretreatment topic shares and national topic flows are the inputs to the instrument. (Table 1; 16 of 18 topics shown)

Identification: A Shift-Share Instrument for Text

- OLS of growth on log provisions is biased by confounders (e.g., a rising industry drives both laws and growth) and reverse causality.
- Novel design: a shift-share (Bartik) instrument for TEXT. States borrow legal language from other states, especially on topics where they have little existing detail.
- Shares: state's pretreatment (1955-64) legislative topic shares, W^k(s,0) / W(s,0).
- Shifts: leave-one-out average log change in the other 49 states' legislating on topic k.
- Instrument Z(s,t) = sum over topics of [pretreatment topic share] x [leave-one-out national topic flow], standardized.
- First stage is NEGATIVE: states starting with LOW detail on a topic legislate most when that topic rises nationally (F = 22.8; effective F = 132.8).
- Identifying assumption: national topic-specific flows are exogenous to a given state (Adao-Kolesar-Morales 2019; Borusyak-Hull-Jaravel 2022); passes placebo, balance, and share-exogeneity checks.

First Stage: The Text Shock Shifts Legislation

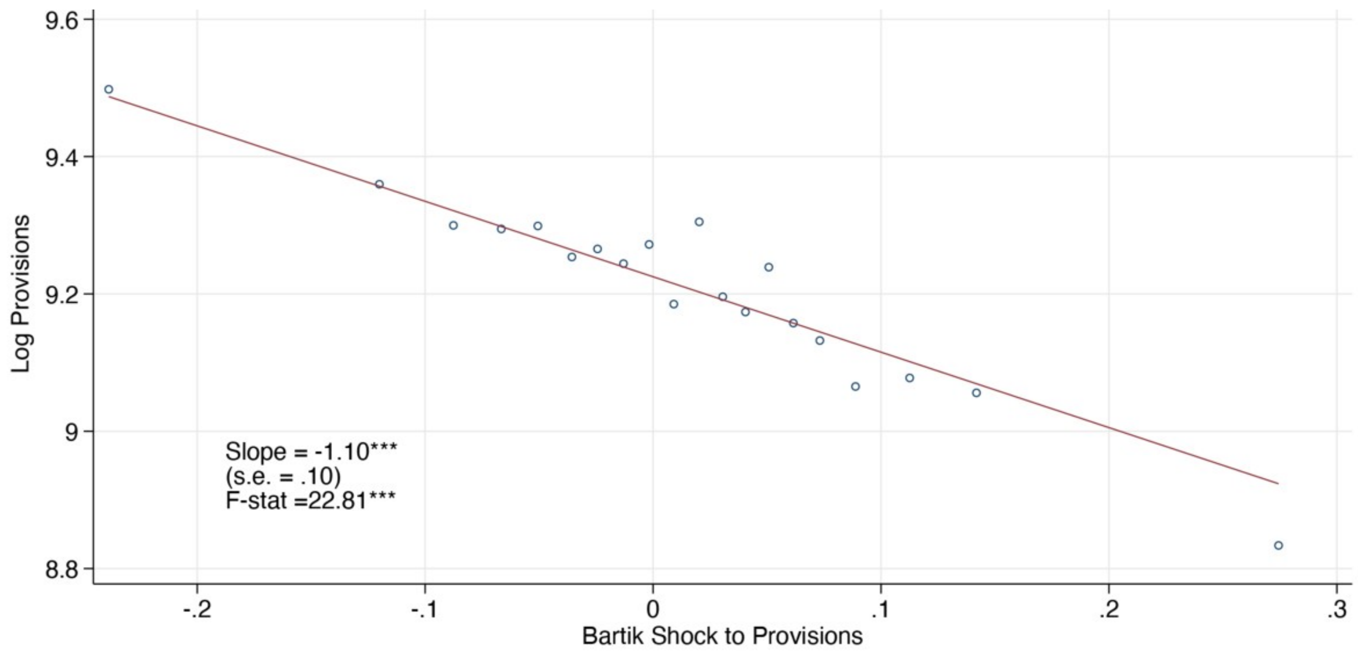

FIG. 2.—First stage: impact of shift-share legislative shock on legislative output. Shown is a binned scatterplot for the first-stage relationship (eq. [3]) between the shift-share instrument (x-axis) and the log number of provisions (y-axis). State and year fixed effects are absorbed.

Note: Binned scatter of the shift-share instrument (x) against log provisions (y), absorbing state and year fixed effects. Negative slope (-1.10, F = 22.8): a positive national topic shock raises legislating most where a state began with low detail on that topic. (Figure 2)

Main Result: More Legislation Causes More Growth

TABLE 3
Effect of Legislative Output on Economic Growth (2SLS)

	Effect on Growth Rate per Capita						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Legislative output	.0182**	.0168*	.0152**	.0134*	.0116*	.0222**	.0094*
	(.00903)	(.00863)	(.00704)	(.00687)	(.00602)	(.0106)	(.00507)
First-stage							
F-statistic	22.86	22.19	23.11	22.92	44.51	19.69	27.30
Observations	1,182	1,182	1,182	1,182	1,134	1,182	1,086
Time fixed effects	X	X	X	X	X	X	X
State fixed effects	X	X	X	X	X	X	X
State trends		X					X
Economic variables × time			X				X
Sector shares × time				X			X
Demographic variables × time					X		X
Topic shares						X	X
Lagged government expenditures							X
Lagged dependent variable							X

Note.—Results for the 2SLS model (second stage [1] and first stage [3]). All specifications include state and biennium fixed effects. Column 2 adds state-specific linear trends. Column 3 adds a set of preperiod economic covariates interacted with biennium effects (initial growth, initial GSP, and initial GSP per capita). Column 4 controls for initial sector shares interacted by biennium, and col. 5 adds demographic characteristics (share of urban, foreign, and population) measured in the pretreatment period interacted with biennium fixed effects. Column 6 includes topic share controls. Column 7 includes all covariates and adds lagged government expenditures and the lagged dependent variable. Descriptions of the covariates with data sources are shown in table A.1. Standard errors are clustered by state.

* $p < .10$.

** $p < .05$.

Note: 2SLS effect of legislative output on real per-capita GDP growth: positive and significant across specifications (state/year FE, state trends, preperiod economic, sector, and demographic controls, topic shares, lags). A 10% instrumented rise in legislation raises growth by about 0.1-0.2 pp (mean 3.1 pp) -- comparable to a 0.1 pp rise in net-of-tax government spending. (Table 3)

Mechanism: Incomplete Contracts & Holdup

- Legislation is an incomplete 'social contract'. More complete laws reduce ex post holdup, encouraging ex ante relationship-specific investment (Williamson; Grossman-Hart; Nunn 2007).
- Because the instrument captures borrowed, positively selected policies, the added detail is efficiency-enhancing on average.
- Five predictions, all supported in the data:
  - Economic (not social) regulations drive the effect; fiscal rules also matter (Table 4).
  - Contingent clauses help most; noncontingent clauses can even hurt (Table 5).
  - Concavity: the effect is larger where preexisting legal detail is low.
  - Effect concentrates in sectors with relationship-specific inputs; none where inputs are exchange-traded (Table 6).
  - Effect is largest under high local economic uncertainty / EPU (Table 7).

Contingent Clauses Drive the Effect

TABLE 5
EFFECT OF CONTINGENT AND NONCONTINGENT CLAUSES ON ECONOMIC GROWTH

	Effect on Real GDP Growth Per Capita						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Contingent provisions	.0638***	.0590***					
	(.0226)	(.0215)					
Noncontingent provisions	−.0559**	−.0511**					
	(.0242)	(.0228)					
Contingent − noncontingent			.0752***	.0697***	.0501**	.0379**	.0773***
			(.0242)	(.0229)	(.0219)	(.0158)	(.0219)
First-stage F-statistic	22.27	36.82	22.83	36.60	15.13	31.68	23.86
Observations	1,182	1,182	1,182	1,182	1,182	1,182	1,134
Time fixed effects	X	X	X	X	X	X	X
State fixed effects	X	X	X	X	X	X	X
State trends		X		X			
Economic variables × time					X		
Sector shares × time						X	
Demographic variables × time							X

NOTE.—Results for the 2SLS model of contingencies. Columns 1 and 2 show results for contingent and noncontingent clauses together (second stage [7] and first stages [8] and [9]), adding state-specific trends in the second column. Columns 3–7 show the results for the difference between contingent and noncontingent clauses (second stage [10] and first stages [11]). Column 4 adds state-specific trends, col. 5 adds preperiod economic variables interacted by year, col. 6 interacts initial sector shares by biennium, and col. 7 shows initial demographic characteristics interacted by biennium. All specifications include controls for state and biennium fixed effects.
** $p < .05$.
*** $p < .01$.

Note: Contingent vs. noncontingent provisions. Contingent clauses have a large positive effect on growth (about 3-4x that of total provisions), while noncontingent clauses are negative; the contingent-minus-noncontingent effect is robustly positive. Consistent with completeness and holdup reduction as the mechanism. (Table 5)

Takeaways

- At the margin, more state legislation causes faster economic growth in the US, 1965-2012.
- Mechanism: more complete laws -- especially contingent clauses -- reduce legal uncertainty and holdup, raising relationship-specific investment; strongest where detail is low, inputs are specific, and uncertainty is high.
- Method 1: an NLP measure of legislative output (parsed provisions + LDA topics) built from raw statute text.
- Method 2: a text-based shift-share instrument exploiting cross-state legal borrowing -- portable to diffusion of technology into patents, source code between projects, or narratives on social media.
- Caveats: a local (marginal) effect in a US setting where competition and learning favor beneficial laws; spillovers and external validity left for future work.